

Applications of Differentiation Trigonometric

Note: the large number sketches should allow the alternative of machine sketching of some graphs, followed by algebraic explanation of the features.

- Find the gradient of the tangent to each of the following curves at the point indicated:
 - $y = \cos x$ at $x = \frac{\pi}{6}$
 - $y = \sin x$ at $x = \frac{\pi}{4}$
 - $y = \tan x$ at $x = 0$
 - $y = \cos 2x$ at $x = \frac{\pi}{4}$
 - $y = \sin \frac{x}{2}$ at $x = \frac{2\pi}{3}$
 - $y = \tan 2x$ at $x = \frac{\pi}{6}$
- Find the equation of the tangent at the given point on each of the following curve
 - $y = \cos 2x$ at $(\frac{\pi}{4}, 0)$
 - $y = \sin 2x$ at $(\frac{\pi}{3}, \frac{\sqrt{3}}{2})$
 - $y = x \sin x$ at $(\pi, 0)$
- Find the equations of the tangent and normal to $y = \sin^2 x$ at the point where $x = \frac{\pi}{4}$.
 - If the tangent meets the x -axis at P and the normal meets the y -axis at Q , find the area of $\triangle OPQ$, where O is the origin.
- Find, in the domain $0 \leq x \leq 2\pi$, the x -coordinates of the points on each of the following curves where the gradient of the tangent is zero.
 - $y = 2 \sin x$
 - $y = 2 \cos x + x$
 - $y = 2 \sin x + \sqrt{3}x$
 - $y = e^{\sin x}$
- Find the equations of the tangent and normal to the curve $y = 2 \sin x - \cos 2x$ at $(\frac{\pi}{6}, \frac{1}{2})$.
- Find the first and second derivatives of $y = \cos x + \sqrt{3} \sin x$
 - Find the stationary points in the domain $0 \leq x \leq 2\pi$ and use the second derivative to determine their nature.
 - Find the points inflexion.
 - Hence sketch the curve for $0 \leq x \leq 2\pi$.
- Repeat the previous question for $y = \cos x - \sin x$. Verify your results by sketching $y = \cos x$ and $y = -\sin x$ on the same diagram and then sketching $y = \cos x - \sin x$ by addition of heights. (notice that the final graph is a sine wave shifted sideways.)
- Find the first and second derivatives of $y = x + \sin x$
 - Find the stationary points in the domain $-2\pi < x < 2\pi$ and determine their nature.
 - Find the points of inflexion for $-2\pi < x < 2\pi$. *Sketch the graph*
- Repeat the previous question for $y = x - \cos x$.
- Find the first and second derivatives of $y = 2 \sin x + \cos 2x$
 - Show that $y' = 0$ when $\cos x = 0$ or $\sin x = \frac{1}{2}$. (You will need to use the formula $\sin 2x = 2 \sin x \cos x$.)
 - Hence find the stationary points in the interval $-\pi \leq x \leq \pi$ and determine their nature.
 - Sketch the curve for $-\pi \leq x \leq \pi$ using this information.
- Repeat the previous question for $y = 2 \cos x + \sin 2x$. (You will need to use the formula $\cos 2x = 1 - 2 \sin^2 x$ to find the stationary points.)
- Find the first and second derivatives of $y = e^{-x} \cos x$. (Note that this function models damped oscillations.)
 - Find the stationary points for $-\pi \leq x \leq \pi$ and determine their nature.
 - Find the points of inflexion for $-\pi \leq x \leq \pi$.
 - Hence sketch the curve for $-\pi \leq x \leq \pi$.
- Repeat the previous question for $y = e^x \sin x$. (Note that this function models oscillations such as feedback loops which grow exponentially.)

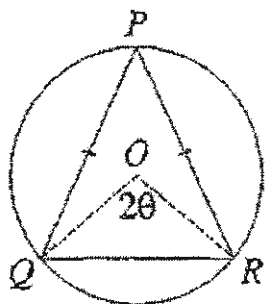
14. The angle θ between two radii OP and OQ of a circle of radius 6 cm is increasing at the rate of 0.1 radians per minute.

- Show that the area of sector OPQ is increasing at the rate of $1.8\text{ cm}^2/\text{min}$
- Find the rate at which the area of ΔOPQ is increasing at the instant when $\theta = \frac{\pi}{4}$.
- Find the value of θ for which the rate of increasing of the area of the segment cut off by the chord PQ is at its maximum.

15. An isosceles triangle has equal sides of length 10 cm . The angle θ between these equal sides is increasing at the rate of 3° per minute. Show that the area of the triangle is increasing at $\frac{5\sqrt{3}\pi}{12}\text{ cm}^2$ per minute at the instant when $\theta = 30^\circ$.

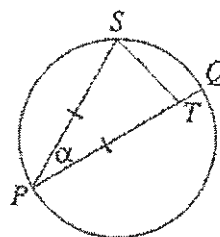
16. A rotating light L is situated at sea 180 metres from the nearest point P on a straight shoreline. The light rotates through one revolution every 10 seconds. Show that the rate at which a ray of light moves along the shore at a point 300 metres from P is $136\pi\text{ m/s}$.

17. ΔPQR is an isosceles triangle inscribed in a circle with centre O of radius one unit, as shown in the diagram. Let $\angle QOR = 2\theta$, where θ is acute.



- Show that the area A of ΔPQR is given by $A = \sin \theta (\cos \theta + 1)$.
- Hence show that, as θ varies, ΔPQR has its maximum possible area when it is equilateral.

18. PQ is a diameter of given circle and S is a point on the circumference. T is the point on PQ such that $PS = PT$. Let $\angle SPT = \alpha$.



- Show that the area A of ΔSPT is $A = \frac{1}{2}d^2 \cos^2 \alpha \sin \alpha$, where d is the diameter of the circle.
- Hence show that the maximum area of ΔSPT as S varies on the circle is $\frac{1}{9}d^2\sqrt{3}\text{ units}^2$

19. A straight line passes through the point $(2,1)$ and has positive x - and y -intercepts at P and Q respectively. Suppose $\angle OPQ = \alpha$, where O is the origin.

- Explain why the line has gradient $-\tan \alpha$
- Find the x - and y -intercepts in terms of α .
- Show that the area of ΔOPQ is given by $A = \frac{(2 \tan \alpha + 1)^2}{2 \tan \alpha}$.
- Hence show that this area is minimised when $\tan \alpha = \frac{1}{2}$.

20. Find the maximum and minimum values of the expression $\frac{2 - \sin \theta}{\cos \theta}$ for $0 \leq \theta \leq \frac{\pi}{4}$, and state the values of θ for which they occur.

21. Find any stationary points and inflexions, then sketch each curve for $0 \leq x \leq 2\pi$:

- $y = 2 \sin x + x$
- $y = 2x - \tan x$

22. Find any stationary points and any other important features, then sketch for $0 \leq x \leq 2\pi$:

- $y = \sin^2 x + \cos x$
- $y = \sin^3 x \cos x$
- $y = \tan^2 x - 2 \tan x$

23. Find the functions v and $\ddot{x}$ for a particle P moving according to $x = 2 \sin \pi t$.

- Show that P is at the origin when $t = 1$, and find its velocity and acceleration then.
- In what direction is the particle:
 - Moving when $t = \frac{1}{3}$?
 - Accelerating when $t = \frac{1}{3}$?

24. A particle is moving according to $x = 4 \cos \frac{\pi}{4} t$, in units of metres and seconds.

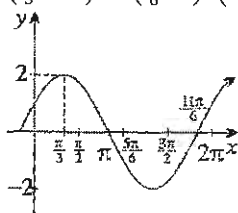
- Find v and $\ddot{x}$, and sketch graphs of x , v and $\ddot{x}$ against t , for $0 \leq t \leq 8$.
- What are the particle's maximum displacement, velocity and acceleration, and when, during the first 8 seconds, do they occur?
- How far does it travel during the first 20 seconds, and what is its average speed?
- When, during the first 8 seconds, is
 - $x = 2$,
 - $x < 2$?
- When, during the first 8 seconds, is
 - $v = \frac{\pi}{2}$,
 - $v > \frac{\pi}{2}$?

25. A particle is oscillating on a spring so that its height is $x = 6 \sin 2t$ cm at time t seconds.

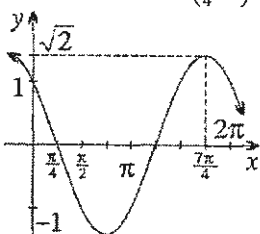
- Find v and $\ddot{x}$ as function of t , and sketch graphs of x , v and $\ddot{x}$, for $0 \leq t \leq 2\pi$.
- Show that $\ddot{x} = -kx$, for some constant k , and find k .
- When, during the first π seconds, is the particle:
 - At the origin,
 - Stationary,
 - Moving with zero acceleration?
- When, during the first π seconds, is the particle:
 - Below the origin
 - Moving downwards
 - Accelerating downwards?
- Find the first time the particle has
 - Displacement $x = 3$,
 - Speed $|v| = 6$.

Answer:

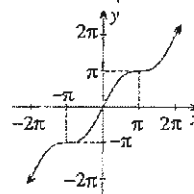
- a. $-\frac{1}{2}$; b. $\frac{1}{\sqrt{2}}$; c. 1; d. -2; e. $\frac{1}{4}$; f. 8
- a. $y = -2x + \frac{\pi}{2}$; b. $x + y + \frac{\pi}{3} + \frac{\sqrt{3}}{2}$; c. $y = -\pi x + \pi^2$
- a. $x - y = \frac{\pi}{4} - \frac{1}{2}$, $x + y = \frac{\pi}{4} + \frac{1}{2}$; b. $\frac{\pi^2 - 4}{32}$ units²
- a. $\frac{\pi}{2}, \frac{3\pi}{2}$; b. $\frac{\pi}{6}, \frac{5\pi}{6}$; c. $\frac{5\pi}{6}, \frac{7\pi}{6}$; d. $\frac{\pi}{2}, \frac{3\pi}{2}$
- $12\sqrt{3}x - 6y = 2\sqrt{3}\pi - 3$, $6x + 12\sqrt{3}y = \pi + 6\sqrt{3}$
- a. $y' = -\sin x + \sqrt{3} \cos x$, $y'' = -\cos x - \sqrt{3} \sin x$; b. maximum turning point $(\frac{\pi}{3}, 2)$, minimum turning point $(\frac{4\pi}{3}, -2)$; c. $(\frac{5\pi}{6}, 0)$, $(\frac{11\pi}{6}, 0)$; d. see below



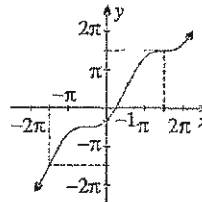
- $y' = -\sin x - \cos x$, $y'' = -\cos x + \sin x$, minimum turning point $(\frac{3\pi}{4}, -\sqrt{2})$, maximum turning point $(\frac{7\pi}{4}, \sqrt{2})$, points of inflexion $(\frac{\pi}{4}, 0)$, $(\frac{5\pi}{4}, 0)$.



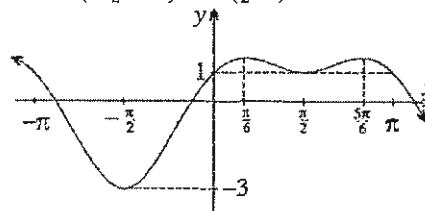
- a. $y' = 1 + \cos x$, $y'' = -\sin x$; b. $(-\pi, -\pi)$ and (π, π) are horizontal points of inflexion; c. $(0, 0)$; d. see below.



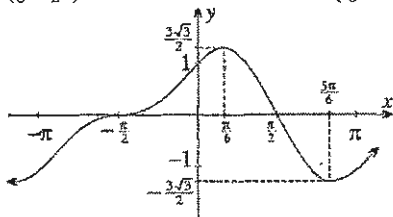
- $y' = 1 + \sin x$, $y'' = \cos x$, horizontal points of inflexion $(-\frac{\pi}{2}, -\frac{\pi}{2})$, $(\frac{3\pi}{2}, \frac{3\pi}{2})$, points of inflexion $(-\frac{3\pi}{2}, -\frac{3\pi}{2})$, $(\frac{\pi}{2}, \frac{\pi}{2})$



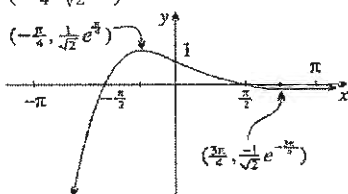
- a. $y' = 2 \cos x - 2 \sin 2x$, $y'' = -2 \sin x - 4 \cos 2x$; c. maximum turning points $(\frac{\pi}{6}, \frac{3}{2})$ and $(\frac{5\pi}{6}, \frac{3}{2})$, minimum turning points $(-\frac{\pi}{2}, -3)$ and $(\frac{\pi}{2}, 1)$; d. see below



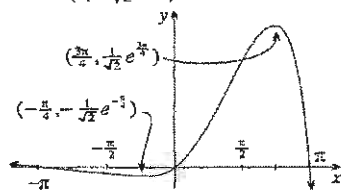
11. horizontal point of inflexion $(-\frac{\pi}{2}, 0)$, maximum turning point $(\frac{\pi}{6}, \frac{3\sqrt{3}}{2})$, minimum turning point $(\frac{5\pi}{6}, -\frac{3\sqrt{3}}{2})$



12. a. $y' = -e^{-x}(\cos x + \sin x)$, $y'' = 2e^{-x} \sin x$; b. minimum turning point $(\frac{3\pi}{4}, -\frac{1}{\sqrt{2}}e^{-\frac{3\pi}{4}})$, maximum turning point $(-\frac{\pi}{4}, \frac{1}{\sqrt{2}}e^{-\frac{\pi}{4}})$; c. $(-\pi, -e^{-\pi})$, $(0, 1)$, $(\pi, -e^{-\pi})$; d. see below



13. minimum turning point $(-\frac{\pi}{4}, -\frac{1}{\sqrt{2}}e^{-\frac{\pi}{4}})$, maximum turning point $(\frac{3\pi}{4}, \frac{1}{\sqrt{2}}e^{-\frac{3\pi}{4}})$, points of inflexion $(-\frac{\pi}{2}, -e^{-\frac{\pi}{2}})$, $(\frac{\pi}{2}, e^{-\frac{\pi}{2}})$

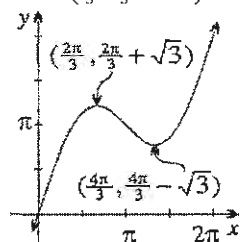


14. b. $\frac{9\sqrt{2}}{10} \text{ cm}^2/\text{min}$; c. $\theta = \pi$

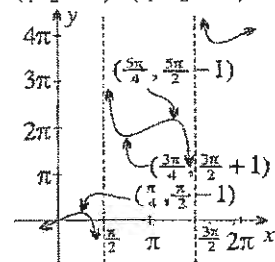
19. a. The angle of inclination is $\pi - \alpha$ and so $m = \tan(\pi - \alpha) = -\tan \alpha$; b. $P = (\frac{1}{\tan \alpha} + 2, 0)$, $Q = (0, 2 \tan \alpha + 1)$

20. minimum $\sqrt{3}$ when $\theta = \frac{\pi}{6}$, maximum 2 when $\theta = 0$

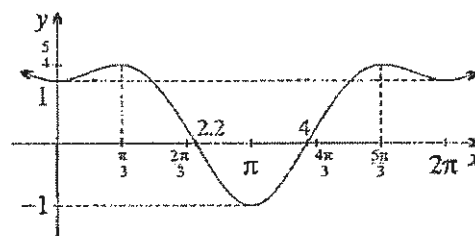
21. a. maximum turning point $(\frac{2\pi}{3}, \frac{2\pi}{3} + \sqrt{3})$, minimum turning point $(\frac{4\pi}{3}, \frac{4\pi}{3} - \sqrt{3})$, inflexion (π, π) ;



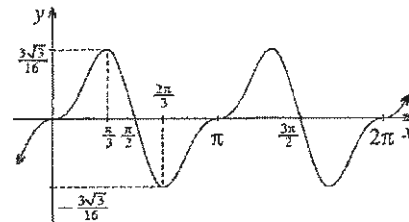
- b. point of inflexion $(\pi, 2\pi)$, minimum turning points $(\frac{3\pi}{4}, \frac{3\pi}{4} + 1)$, $(\frac{7\pi}{4}, \frac{7\pi}{4} + 1)$, maximum turning points $(\frac{\pi}{4}, \frac{\pi}{4} - 1)$, $(\frac{5\pi}{4}, \frac{5\pi}{4} - 1)$



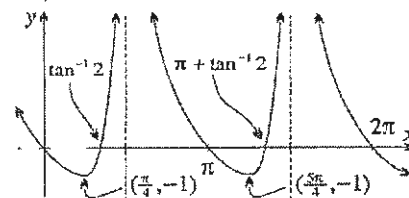
22. a. maximum turning points $(\frac{\pi}{3}, \frac{5}{4})$, $(\frac{5\pi}{3}, \frac{5}{4})$, minimum turning points $(0, 1)$, $(\pi, -1)$, $(2\pi, 1)$, x -intercepts $\pi - \cos^{-1} \frac{\sqrt{5}-1}{2} \doteq 2.2$, $\pi + \cos^{-1} \frac{\sqrt{5}-1}{2} \doteq 4.0$



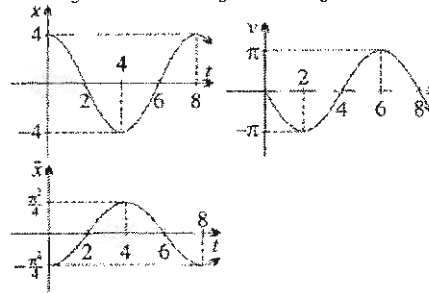
- b. maximum turning points $(\frac{\pi}{3}, \frac{3\sqrt{3}}{16})$, $(\frac{4\pi}{3}, \frac{3\sqrt{3}}{16})$, minimum turning points $(\frac{2\pi}{3}, -\frac{3\sqrt{3}}{16})$, $(\frac{5\pi}{3}, -\frac{3\sqrt{3}}{16})$, horizontal points of inflexion $(0, 0)$, $(\pi, 0)$, $(2\pi, 0)$



- c. minimum turning points $(\frac{\pi}{4}, -1)$, $(\frac{5\pi}{4}, -1)$, vertical asymptotes $x = \frac{\pi}{2}$, $x = \frac{3\pi}{2}$, x -intercepts $0, \pi, 2\pi, \tan^{-1} 2 \doteq 1.1$, $\pi + \tan^{-1} 2 \doteq 4.25$



23. $v = 2\pi \cos \pi t$, $\ddot{x} = -2\pi^2 \sin \pi t$; a. When $t = 1$, $x = 0$, $v = -2\pi$ and $\ddot{x} = 0$; b. i. right ($v = \pi$), ii. left ($\ddot{x} = -\pi^2\sqrt{3}$)
24. a. $x = 4 \cos \frac{\pi}{4} t$, $v = -\pi \sin \frac{\pi}{4} t$, $\ddot{x} = -\frac{1}{4}\pi^2 \cos \frac{\pi}{4} t$ (graph see below); b. maximum displacement $x = 4$ when $t = 0$ and $t = 8$, maximum velocity $\pi \text{ m/s}$ when $t = 6$, maximum acceleration: $\frac{1}{4}\pi^2 \text{ m/s}^2$ when $t = 4$; c. 40 metres, 2 m/s ; d. i. after $1\frac{1}{3}$ and $6\frac{2}{3}$ seconds, ii. $1\frac{1}{3} < t < 6\frac{2}{3}$; e. i. after $4\frac{2}{3}$ and $7\frac{1}{3}$ seconds, ii. $4\frac{2}{3} < t < 7\frac{1}{3}$



25. a. $x = 6 \sin 2t$, $v = 12 \cos 2t$, $\ddot{x} = -24 \sin 2t$ (graph see below); b. $\ddot{x} = -4x$; c. i. $x = 0$ when $t = 0, \frac{\pi}{2}$ or π , ii. $v = 0$ when $t = \frac{\pi}{4}$ or $\frac{3\pi}{4}$, iii. Same as i.; d. i. $x < 0$ when $\frac{\pi}{2} < t < \pi$, $v < 0$ when $\frac{\pi}{4} < t < \frac{3\pi}{4}$, iii. $\ddot{x} < 0$ when $0 < t < \frac{\pi}{2}$; e. i. $t = \frac{\pi}{12}$, ii. $t = \frac{\pi}{6}$

